SERVICENOW SYSTEM ADMINISTRATOR · NAAN MUDHALVAN (SMARTBRIDGE)

Proposed Solution — Script-Controlled ACL for Institution Details

Phase 3 · Project Design Phase

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This document states the proposed solution for the Script-Controlled ACL capstone: a native ServiceNow configuration on PDI dev443713 that restricts access to Institution Details records based on both the user's role and the Branch field value. It is presented as a parameter-by-parameter table covering the problem, the solution, its novelty, impact, model and scalability.

Parameter	Description
Problem Statement	On the u_institution_details table, any user who can reach the table can read and edit every record irrespective of branch. An EEE branch user can view and modify ECE and CSE student records, which breaks data privacy and record integrity. There is no field-value boundary and no per-operation control.
Proposed Solution	Configure four record-level ACLs on u_institution_details, one per operation. READ requires role bb1, is set to Advanced, carries the data condition Branch is EEE, and runs a GlideScript using gs.hasRole that returns true only for admin or bb1. CREATE requires bb2, WRITE requires bb3, DELETE requires bb4 (no condition or script). A test user, EEE User, holds bb1-bb4. Editing ACLs needed security_admin elevation, and the auto-generated u_institution_details_user role was removed so each ACL relies only on its bb role.
Novelty / Uniqueness	Access is decided by BOTH the user's role AND the Branch value on each record, combining a declarative role requirement, a data condition, and a server-side script inside a single Advanced ACL rather than one blanket table role. It uses only native platform security (sys_security_acl, GlideScript), with no external app, framework or database.
Social Impact	Protects student personal data (name, email, phone) branch by branch, enforces least-privilege access, prevents accidental cross-branch edits, and upholds the institution's data-privacy expectations for its student records.
Business / Institutional Model	Built entirely as native configuration on a free ServiceNow Personal Developer Instance, so there is no licensing or infrastructure cost beyond the institution's existing ServiceNow. The pattern is reusable: any department or branch can adopt the same role-plus-condition-plus-script ACL design.

Parameter	Description
Scalability of the Solution	The pattern extends to additional branches by adding roles and matching data conditions, applies to other tables and choice fields, and scales from a single PDI to a production enterprise instance because ACL evaluation is part of the platform's core security engine.